

CloudNova Project – Part 2 Report

Group number – 1

1. Introduction

This report presents the monitoring, security, and log-management configuration for the CloudNova AWS environment. Part 2 builds on the infrastructure from Part 1 by implementing CloudWatch, CloudTrail, GuardDuty, Trusted Advisor, and S3 lifecycle policies to enhance visibility, threat detection, and operational reliability. These services collectively help maintain a secure, well-monitored cloud environment aligned with AWS best practices.

2. CloudWatch Logs, Metrics, and Alarms

2.1 CloudTrail Overview

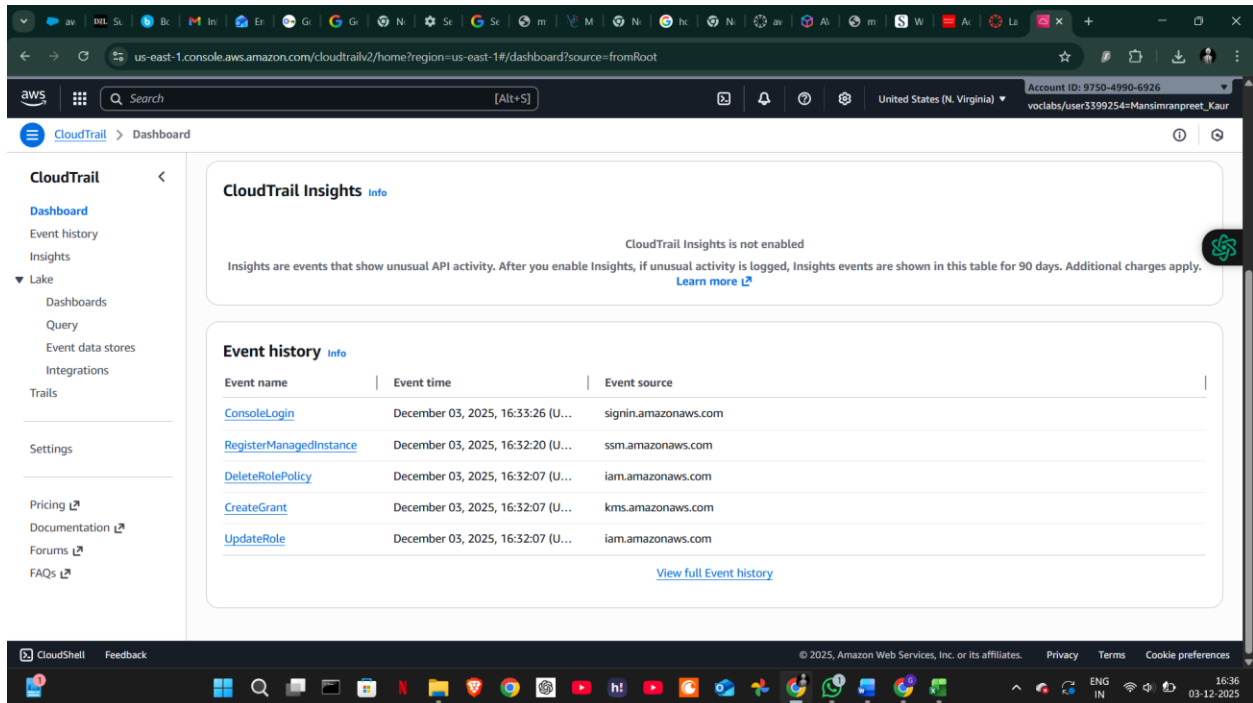
AWS CloudTrail functions as a *security camera* for the AWS environment by recording all user and service activity.

CloudTrail captures:

- Console logins
- API calls
- Resource creation and deletion
- Permission and IAM policy changes
- Security Group modifications
- EC2 start/stop actions
- S3 bucket updates

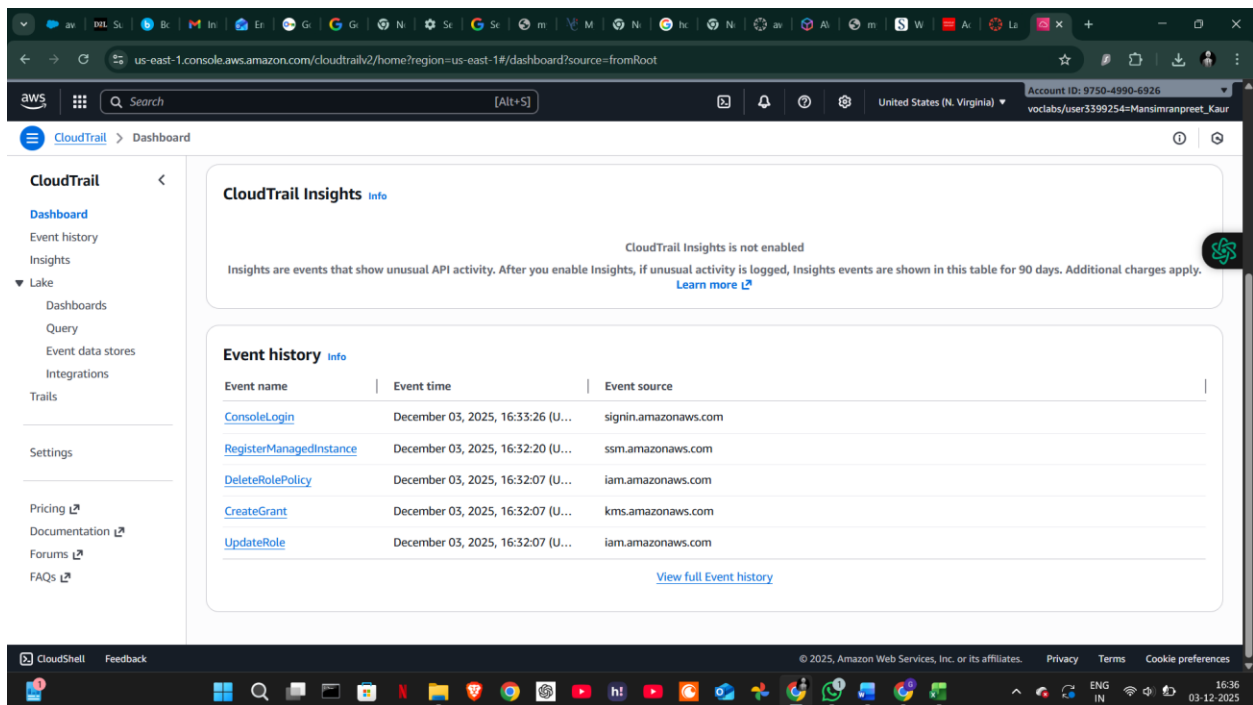
Example CloudTrail Events

- *ConsoleLogin* → user logged in
- *RunInstances* → EC2 instance launched



Log Storage

- Default: S3 bucket
- Optional: CloudWatch Logs (if enabled)



2.2 What is Amazon CloudWatch?

CloudWatch is a monitoring service used to observe application performance and resource usage. It collects logs, tracks metrics, and triggers alarms based on thresholds.

CloudWatch monitors:

- EC2 CPU usage
- RAM (with agent installed)
- Network traffic
- Disk read/write operations
- Application/system logs
- Custom metrics

2.3 CloudWatch Metrics

Metrics are numerical measurements collected every 1 or 5 minutes.

Metrics answer:

“How is my system performing right now?”

EC2 Metrics

- CPU Utilization
- DiskReadBytes / DiskWriteBytes
- StatusCheckFailed

EBS Metrics

- VolumeReadBytes
- VolumeWriteBytes

Custom Metrics (From CloudTrail Metric Filters)

- Failed login attempts
- Unauthorized API actions

2.4 CloudWatch Logs

CloudWatch Logs store text-based logging data such as:

- Application logs
- System logs
- CloudTrail logs
- VPC Flow Logs

Logs answer:

“What exactly happened within my system?”

Tools used:

- Log Groups
- Log Streams
- Logs Insights
- Metric filters

2.5 CloudWatch Alarms

Alarms notify when system performance exceeds or falls below defined thresholds.

Alarms Configured

- CPU alarm → triggered at $\geq 70\%$ CPU utilization
- Disk alarm → based on read/write operations
- Failed login alarm → custom metric from CloudTrail

us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#alarmsV2:

CloudWatch > Alarms

Alarms (3)

Hide Auto Scaling alarms Clear selection Create composite alarm Actions Create alarm

Search Alarm state: Any Alarm type: Any Actions status: Any

Name	State	Last state update (UTC)	Conditions	Actions
high-cpu-utilisation	OK	2025-12-03 21:34:23	CPUUtilization >= 70 for 1 datapoints within 5 minutes	Actions enabled
DiskAlarm-VolumeReadBytes1	Insufficient data	2025-12-03 17:47:57	VolumeReadBytes >= 1000000000 for 1 datapoints within 5 minutes	Actions enabled
FailedLogins	Insufficient data	2025-11-30 05:47:44	ConsoleLoginFailureCount >= 3 for 1 datapoints within 5 minutes	Actions enabled

CloudShell Feedback

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Notifications were sent via Amazon SNS email alerts.

us-east-1.console.aws.amazon.com/sns/v3/home?region=us-east-1#/subscriptions

Amazon SNS > Subscriptions

New Feature Amazon SNS now supports High Throughput FIFO topics. Learn more

Subscriptions (2)

Edit Delete Request confirmation Confirm subscription Create subscription

Search

ID	Endpoint	Status	Protocol	Topic
aaf0691c-f9cf-4fe7-8815-5...	singh.gurjot1729@gmail.com	Confirmed	EMAIL	Default_CloudWatch_Alarm...
843fc18e-b2a8-4eec-a62e-...	arn:aws:lambda:us-east-1:9...	Confirmed	LAMBDA	RedshiftSNS

CloudShell Feedback

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3. AWS GuardDuty – Threat Detection

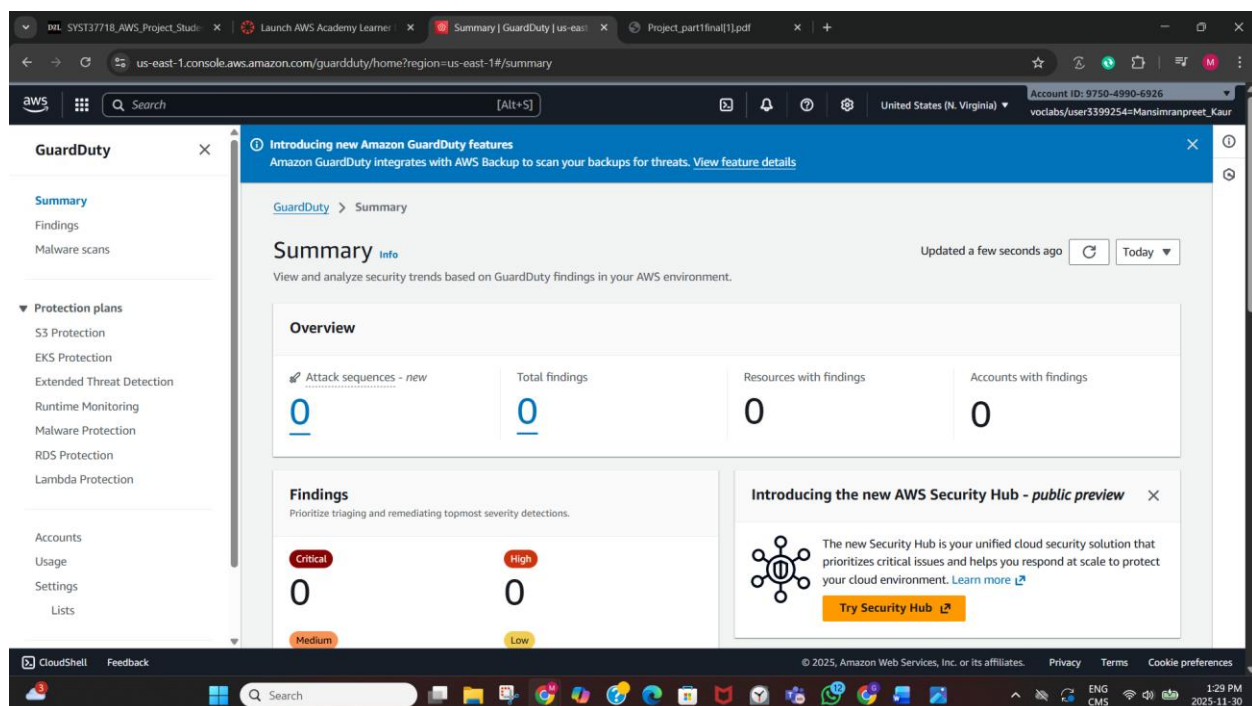
Amazon GuardDuty was enabled to detect malicious or suspicious activity by analyzing:

- CloudTrail logs
- VPC Flow Logs
- DNS query logs

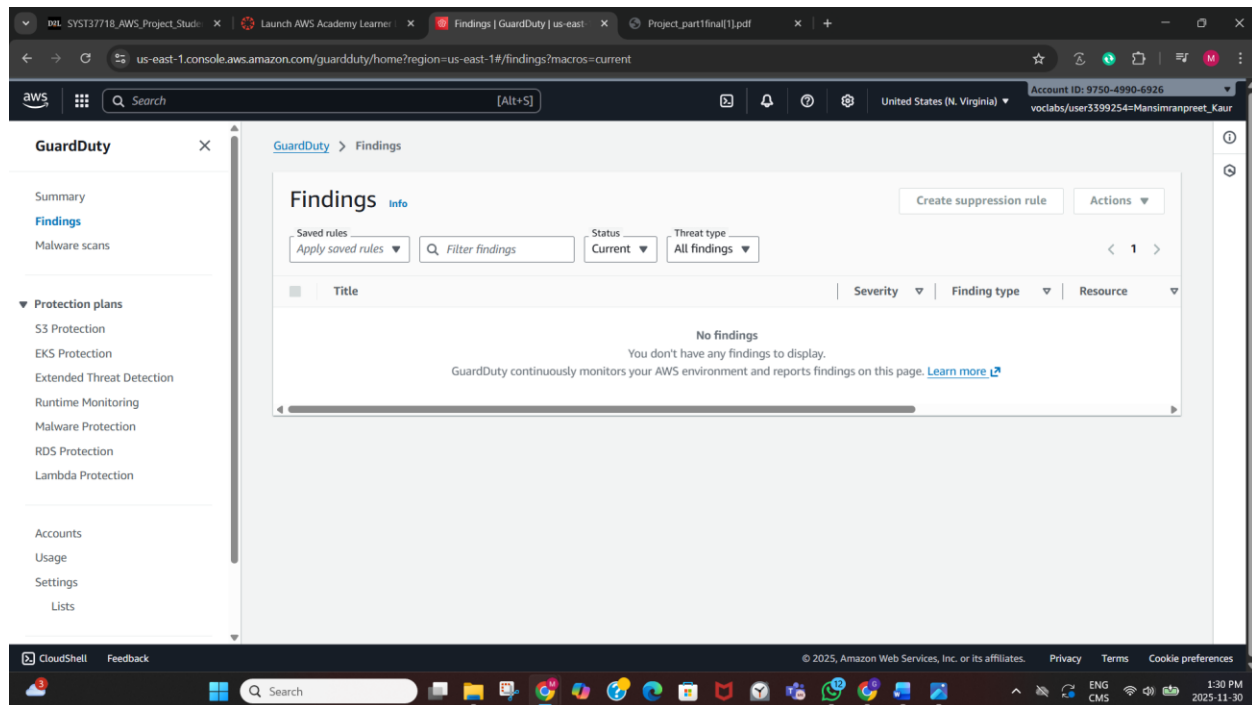
Project Findings

- Attack sequences: 0
- Total findings: 0
- Affected resources: 0
- Accounts with findings: 0

This confirms no abnormal activity occurred during monitoring.



The **Findings** page also reported “No Findings”, confirming that there were no alerts related to unauthorized access, abnormal traffic, or compromised resources.

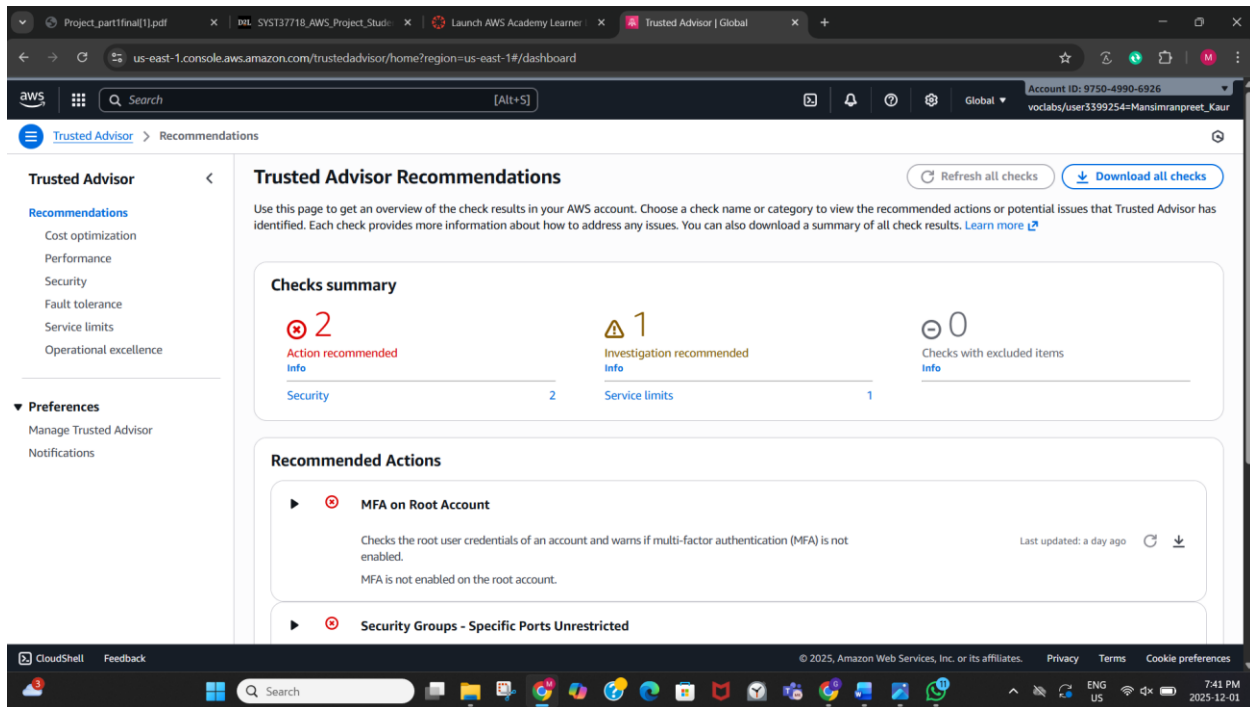


4. Trusted Advisor Findings

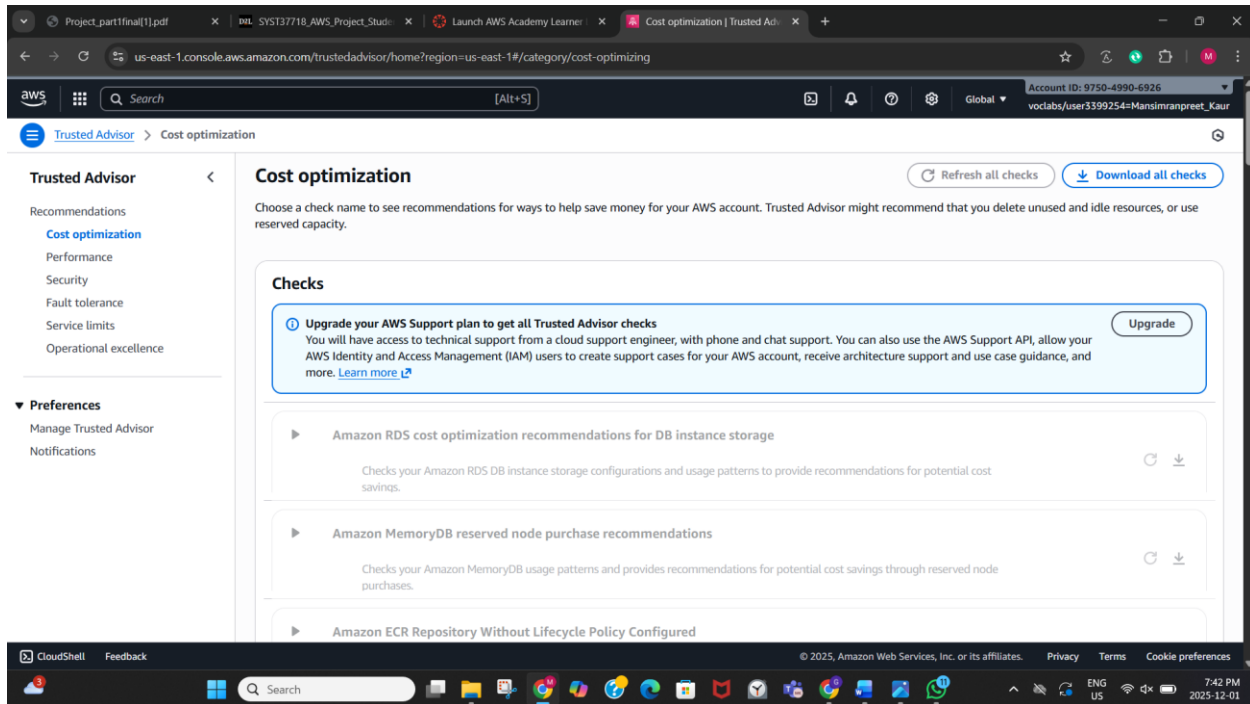
Trusted Advisor evaluates the AWS environment based on cost, performance, security, fault tolerance, and service limits.

Security Issues Identified

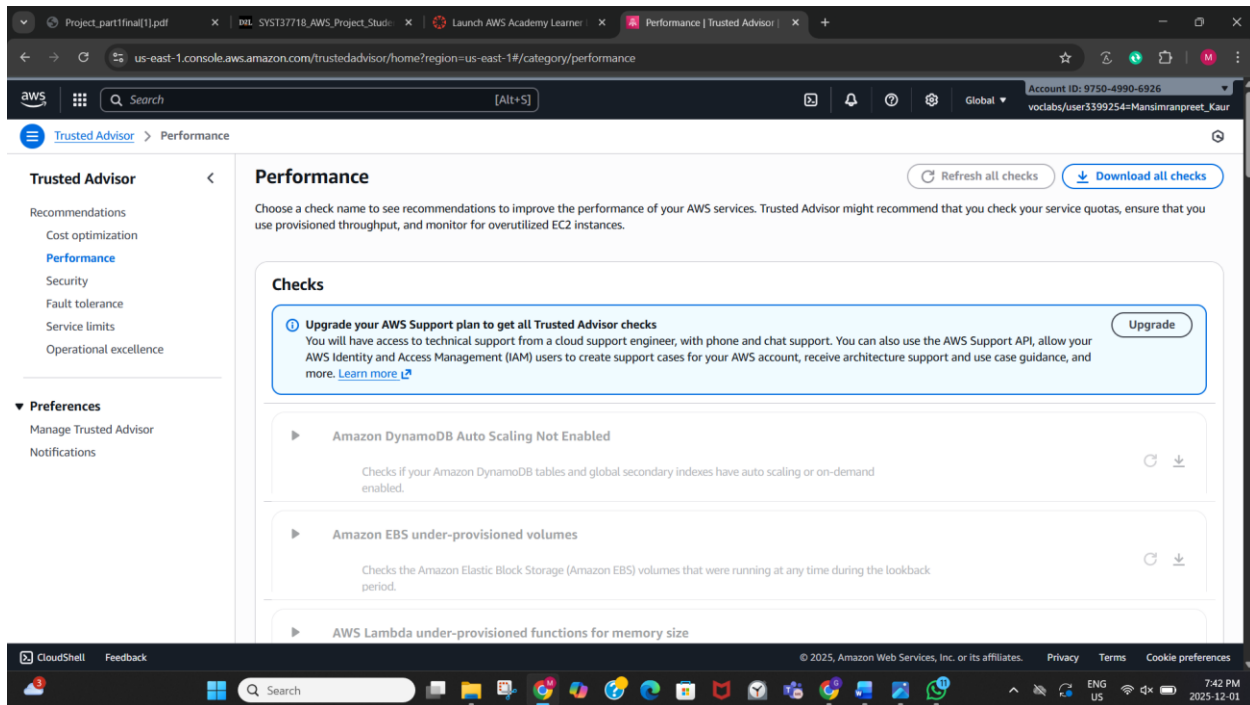
- MFA not enabled on Root Account: Recommended to improve account security
- Security groups allowing unrestricted inbound ports: At least one SG allowed access from 0.0.0.0/0Service Limit Warning
- One warning under *Investigation Recommended*, highlighting that capacity limits may soon be reached.



- **Cost Optimization**



- **Performance**



- **Security**

Under the Security category, Trusted Advisor reports two issues:

- 1. MFA on Root Account – Not Enabled**

Trusted Advisor recommends enabling Multi-Factor Authentication (MFA) on the AWS root user to strengthen account security.

- 2. Security Groups – Specific Ports Unrestricted**

At least one security group allows inbound access from **0.0.0.0/0**, meaning it is open to the entire internet.

Restricting these ports is recommended to reduce exposure to attacks.

These findings highlight common security risks in AWS beginner setups and help ensure better protection of EC2 and VPC resources.

The screenshot shows the AWS Trusted Advisor console in the 'Security' category. The left sidebar lists 'Trusted Advisor' and 'Security' as active sections, with sub-items like 'Recommendations', 'Cost optimization', 'Performance', 'Security', 'Fault tolerance', 'Service limits', and 'Operational excellence'. The main content area is titled 'Security' and includes a 'Refresh all checks' button and a 'Download all checks' button. A message box states: 'Trusted Advisor checks sourced from AWS Security Hub. You can enable Security Hub to manage and improve your security posture. You can then view your Security Hub findings as Trusted Advisor check recommendations below. If you're new to Security Hub or don't see recommendations sourced from Security Hub below, see the documentation.' Below this, the 'Checks' section lists two items: 'MFA on Root Account' (Last updated: a day ago) and 'Security Groups - Specific Ports Unrestricted' (Last updated: 2 minutes ago). The bottom of the screen shows the Windows taskbar and the AWS console footer with the URL 'https://us-east-1.console.aws.amazon.com/trustedadvisor/home?region=us-east-1#/category/security'.

- **Fault Tolerance**

The screenshot shows the AWS Trusted Advisor console in the 'Fault tolerance' category. The left sidebar lists 'Trusted Advisor' and 'Fault tolerance' as active sections, with sub-items like 'Recommendations', 'Cost optimization', 'Performance', 'Security', 'Fault tolerance', 'Service limits', and 'Operational excellence'. The main content area is titled 'Fault tolerance' and includes a 'Refresh all checks' button and a 'Download all checks' button. A message box states: 'Choose a check name to see recommendations to increase the availability and redundancy of your AWS applications. Trusted Advisor might recommend that you use resources such as Auto Scaling, health checks, multiple Availability Zones, and backup capabilities.' Below this, the 'Checks' section lists two items: 'AWS STS global endpoint usage across AWS Regions' (Last updated: 3 minutes ago) and 'Amazon Aurora MySQL cluster backtracking not enabled' (Last updated: 3 minutes ago). A message box also states: 'Upgrade your AWS Support plan to get all Trusted Advisor checks. You will have access to technical support from a cloud support engineer, with phone and chat support. You can also use the AWS Support API, allow your AWS Identity and Access Management (IAM) users to create support cases for your AWS account, receive architecture support and use case guidance, and more. Learn more.' The bottom of the screen shows the Windows taskbar and the AWS console footer with the URL 'https://us-east-1.console.aws.amazon.com/trustedadvisor/home?region=us-east-1#/category/fault-tolerance'.

- **Service Limits**

The screenshot displays the AWS Trusted Advisor console in the 'Service limits' category. The left sidebar shows navigation options: Recommendations, Cost optimization, Performance, Security, Fault tolerance, Service limits (selected), and Operational excellence. Under 'Service limits', there are links for 'Manage Trusted Advisor' and 'Notifications'. The main content area, titled 'Service limits', includes a description: 'Choose a check name to see recommendations for services that use more than 80 percent of a service quota. The check results use values based on a snapshot, so your current usage might vary. Quota and usage data can take up to 24 hours to reflect any changes.' Below this, three checks are listed:

- EC2-VPC Elastic IP Address**: Checks for usage that is more than 80% of the EC2-VPC Elastic IP Address Limit. 1 of 16 items have usage that is more than 80% of the service limit. Last updated: a day ago.
- Auto Scaling Groups**: Checks for usage that is more than 80% of the Auto Scaling Groups Limit. 0 of 17 items have usage that is more than 80% of the service limit. Last updated: 3 minutes ago.
- Auto Scaling Launch Configurations**: Checks for usage that is more than 80% of the Auto Scaling Launch Configurations Limit. 0 of 17 items have usage that is more than 80% of the service limit. Last updated: 3 minutes ago.

Buttons for 'Refresh all checks' and 'Download all checks' are located at the top right of the checks list.

- **Operational Excellence**

The screenshot displays the AWS Trusted Advisor console in the 'Operational excellence' category. The left sidebar shows navigation options: Recommendations, Cost optimization, Performance, Security, Fault tolerance, Service limits, and Operational excellence (selected). Under 'Operational excellence', there are links for 'Manage Trusted Advisor' and 'Notifications'. The main content area, titled 'Operational excellence', includes a description: 'Choose a check name to see recommendations to improve the operational readiness of your AWS resources. Trusted Advisor might recommend that you enable various features and services in AWS to ensure it is ready for operation.' Below this, three checks are listed:

- Upgrade your AWS Support plan to get all Trusted Advisor checks**: You will have access to technical support from a cloud support engineer, with phone and chat support. You can also use the AWS Support API, allow your AWS Identity and Access Management (IAM) users to create support cases for your AWS account, receive architecture support and use case guidance, and more. [Learn more](#). (Upgrade button)
- Amazon Redshift cluster audit logging**: Checks if your Amazon Redshift clusters have database audit logging turned on.
- Amazon VPC Without Flow Logs**: Checks if Amazon VPC Flow Logs are created for a VPC.
- Amazon SNS Topics Not Logging Message Delivery Status**

Buttons for 'Refresh all checks' and 'Download all checks' are located at the top right of the checks list.

These findings help identify common weaknesses in beginner AWS setups and guide improvements.

5. S3 Lifecycle Rules – Log Archiving Simulation

A lifecycle policy was configured on the CloudTrail S3 bucket to simulate automated long-term log management.

Lifecycle Transitions

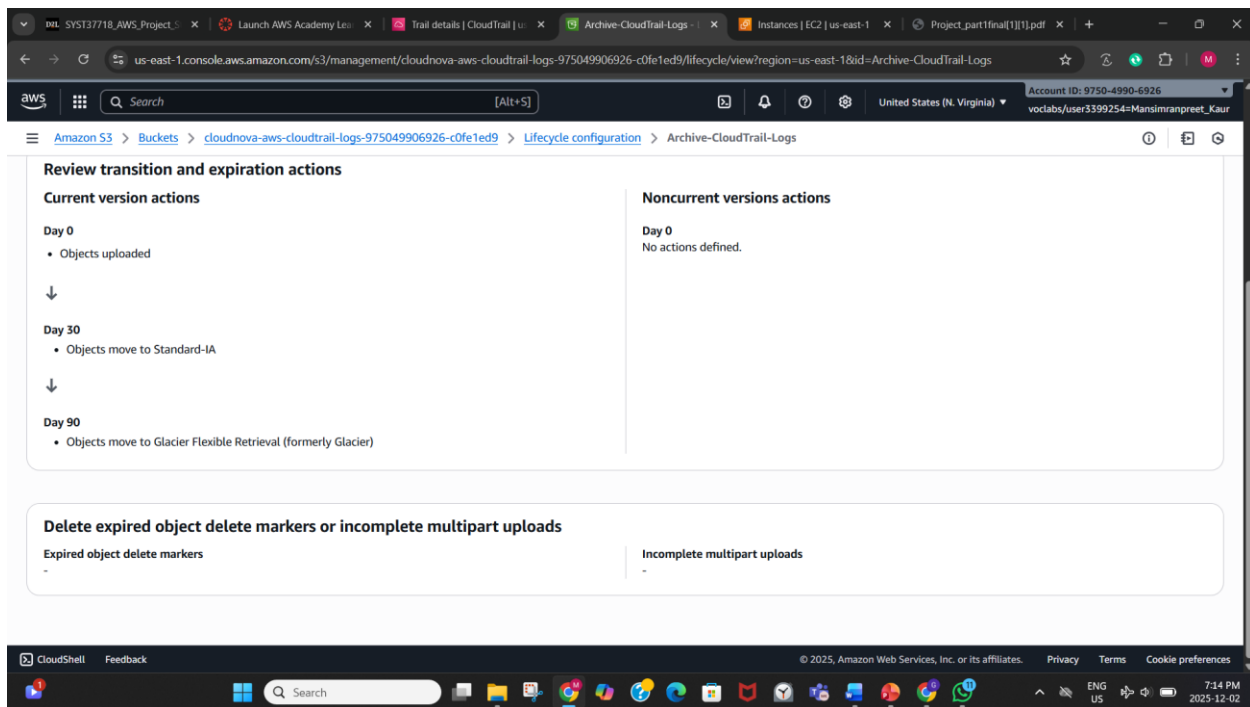
1. After 30 days → Move logs to Standard-IA
 - Suitable for infrequent access
 - Cost-effective while maintaining quick retrieval
2. After 90 days → Move logs to Glacier Flexible Retrieval
 - Best for long-term storage
 - Extremely low-cost archival

No deletion rule was applied, as the project required only archiving simulation.

Purpose of Lifecycle Policy

- Demonstrates automated log retention
- Reduces long-term storage cost
- Supports compliance practices
- Mimics real enterprise log management workflows

The screenshot displays the AWS Management Console interface for the 'Archive-CloudTrail-Logs' bucket. The breadcrumb navigation shows the path: Amazon S3 > Buckets > cloudnova-aws-cloudtrail-logs-975049906926-c0fe1ed9 > Lifecycle configuration > Archive-CloudTrail-Logs. The page title is 'Archive-CloudTrail-Logs' with an 'Info' link. There are 'Edit', 'Delete', and 'Actions' buttons in the top right. The main content is divided into two sections: 'Lifecycle rule configuration' and 'Review transition and expiration actions'. The 'Lifecycle rule configuration' section shows: Lifecycle rule name: Archive-CloudTrail-Logs; Status: Enabled (with a green checkmark); Scope: Entire bucket; Prefix: -; Object tags: -; Minimum object size: - (with a note: 'When no minimum object size is specified, the minimum object size for transitions is determined by the lifecycle configuration. Learn more'); Maximum object size: -. The 'Review transition and expiration actions' section is split into 'Current version actions' and 'Noncurrent versions actions'. Under 'Current version actions', it shows 'Day 0' with 'Objects uploaded' and a downward arrow, and 'Day 30' with the text 'Objects moved to Standard-IA'. The 'Noncurrent versions actions' section shows 'Day 0' with 'No actions defined.' The bottom of the screenshot shows the Windows taskbar with various application icons and the system clock indicating 7:13 PM on 2025-12-02.



6. Summary

Part 2 of the CloudNova project introduced monitoring, auditing, threat detection, and log archiving capabilities. These additions ensure the CloudNova environment is secure, observable, and aligned with real-world cloud operations. CloudWatch and CloudTrail provide visibility, GuardDuty enhances threat detection, Trusted Advisor identifies risks, and S3 lifecycle policies support cost-efficient log retention.